

Viability of blockchain markets

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The Microstructure Exchange
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Motivation

Role of blockchain markets likely to grow

- Tokenisation
 - Gold, Treasuries, Debt, Credit, Private Equity, Stocks can be traded today on blockchains
 - SEC innovation exemption
 - blockchains (Tokenized Securities Venues) exempted from the regulation of exchanges
 - liquidity providers exempted from the regulation of dealers
 - projected to attain \$ 5.5T by 2030 (Citi)
- Stablecoins
 - \$ 310B
- FX cross-border settlement

⇒ important to study

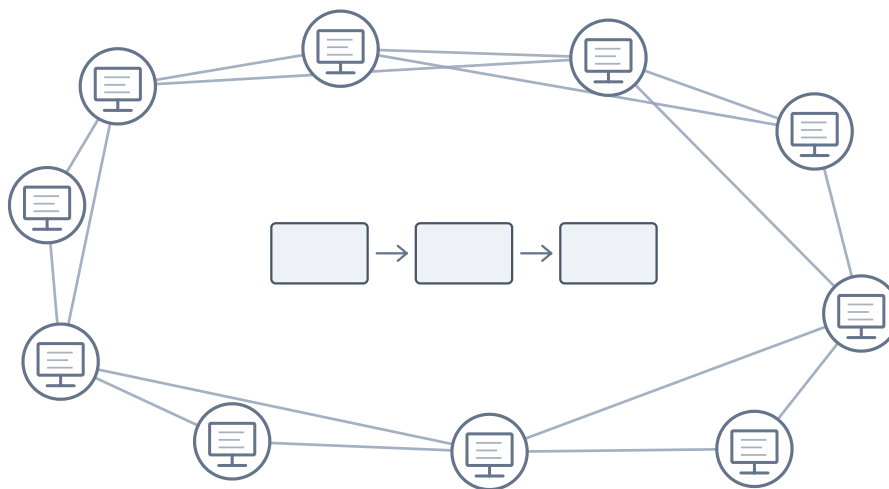
Decentralisation

Traditionally

- Intermediaries (banks, brokers) extract rents, and central authorities (government, regulators) concentrate decision-making power
- Decentralisation: no single central authority, rent is shared

Technological solution: blockchain

- **Digital ledger:** records accounts, balances, and transactions
- **Network of participants** maintains and secures this record using cryptography



Consensus

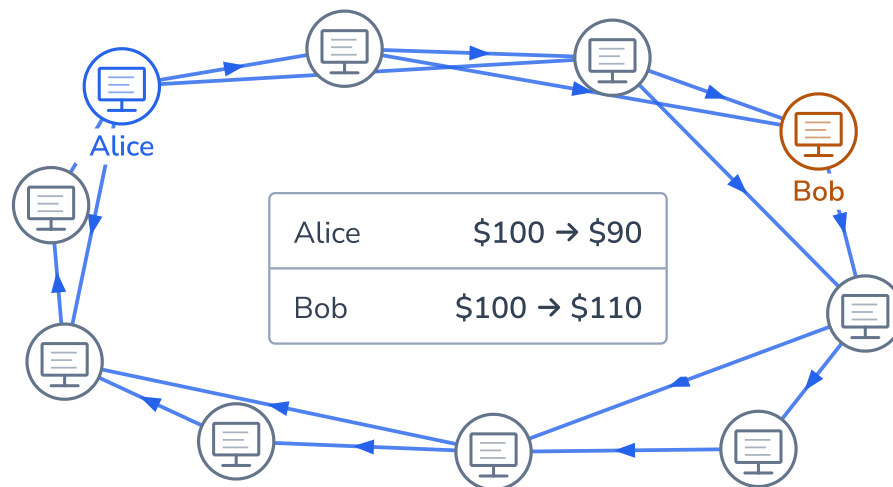
Trust = **consensus**

- update the ledger through **communication and validation**

Example: Alice sends \$10 to Bob

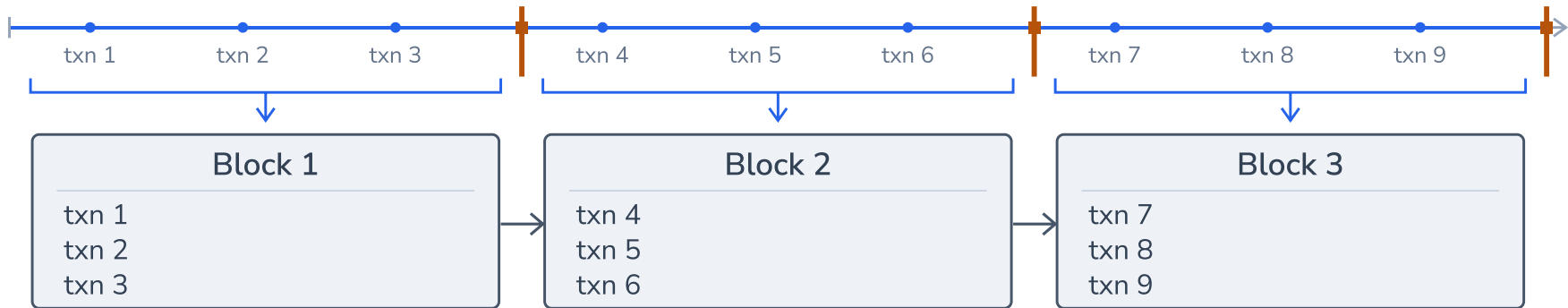
- **Propagate**: Alice signs and broadcasts the transfer
- **Verify**: blockchain participants **validate**
- **Agree**: blockchain participants **vote** and record the transaction across the network

⇒ Decentralisation takes time



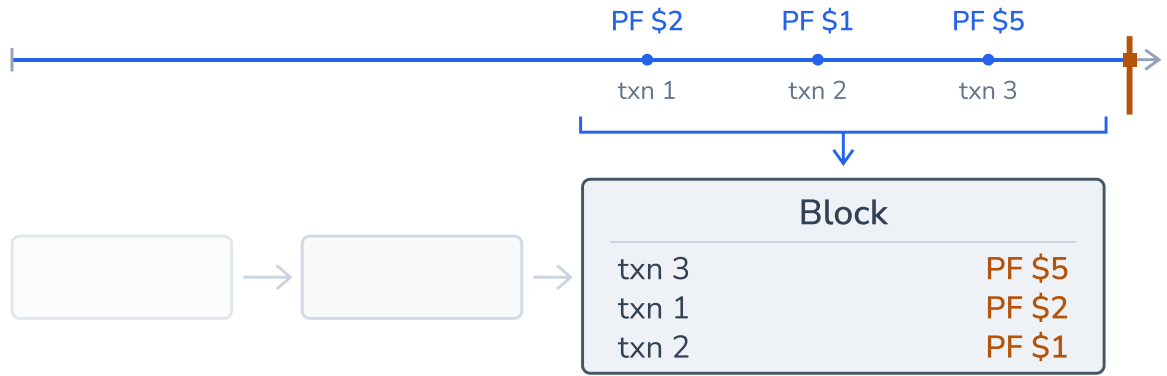
Blocks

- **Solution:** economy of scale: batching to reduce coordination costs
- Transactions are sent continuously, but the market clears periodically (12 seconds on Ethereum)

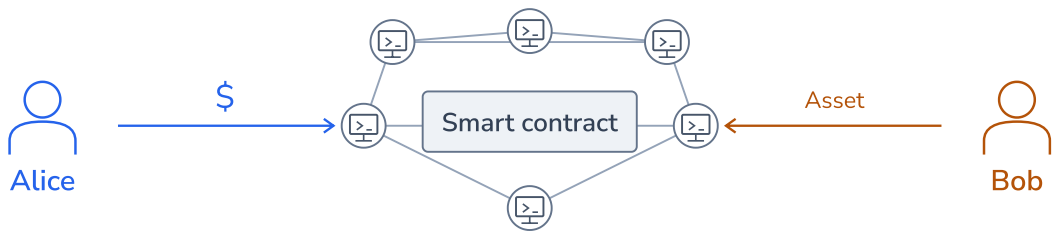


Paid-priority

- Priority fees are tips to bid for earlier execution within a block
- Introduced as an incentive to conduct consensus



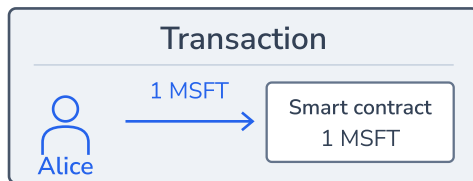
Smart contracts



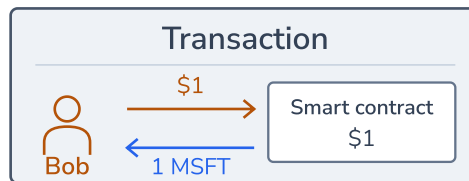
- Smart contract: public immutable code (functions), data (state), and a balance (target of transactions)

Smart contracts in practice

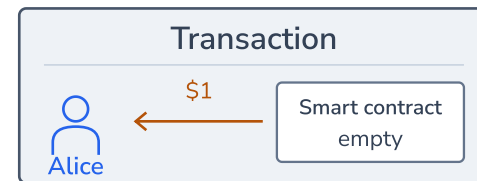
- Example:
 - Smart contract with MSFT and USD reserves
 - Code to exchange one MSFT for one USD



Deposit



Exchange



Withdrawal

- Smart contracts can do more: payments, lending, insurance, derivatives, **trading**
- **Decentralised finance**: collection of financial services that run on the blockchain, offered through smart contracts

Decentralised exchange

- Smart contract for liquidity providers and liquidity takers (traders)
- Liquidity providers supply reserves (MSFT and USDC)
- Traders exchange MSFT for USDC through the smart contract
- Smart contract determines a
 - price schedule
 - price update rule

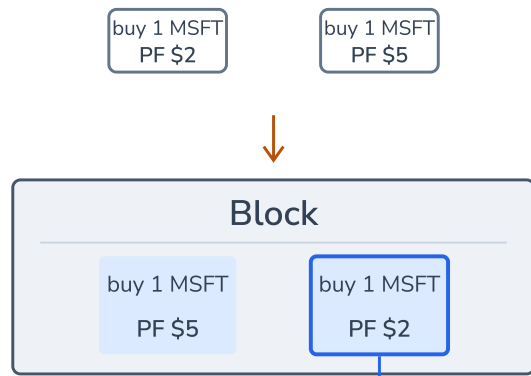


Blockchain trading round

1. LPs commit passive liquidity
price schedule, price update rule



2. Traders submit
transactions + priority fees



3. Block created
trades clear sequentially
in priority order



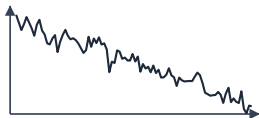
executes against passive liquidity

mechanical price update rule

Are blockchain markets different?

Electronic markets

Continuous trading



Price discovery is **continuous**

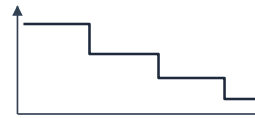
Liquidity provision is **dynamic**

Price impact is **Bayesian**

Time priority

Blockchain markets

Discrete trading



Price discovery is **discrete**

Liquidity provision is **static**

Price impact is **mechanical**

Time + paid-priority

Are these differences good or bad for markets?

Reasons to worry

Blockchain amplifies price moves

- One-block return volatility, on-chain vs off-chain

Asset	Blockchain volatility (bp)	Binance volatility (bp)
ETH/BTC	7.72	6.00
ETH	8.71	6.87
BTC	24.94	17.83
LINK	27.35	13.45
UNI	29.41	18.29
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1INCH	84.11	34.84

Volatility of absolute one-block returns in bp (1 Jan 2021 – 31 Dec 2023)
Data: 8 assets priced both on-chain (Uniswap v3) and on Binance

Contribution

This paper

- Can the blockchain host *viable* markets?
- A model of a **blockchain trading round**

Preview of Result 1

Paid-priority + mechanical price impact = endogenous selection of high-valuation traders in each block

- Trading volumes are endogenous
 - ⇒ traders anticipate that trading large volumes requires high priority fees
 - ⇒ low-valuation traders anticipate large volumes executed ahead of them in the block
 - ⇒ optimally stay out
- Consequences
 - price moves are amplified
 - liquidity dries up

Contribution

Preview of Result 2

Cheaper information impairs prices and liquidity

- The number of competitors is endogenous
- Cheaper information $\implies$ more traders willing to pay the cost of information
- more traders competing for paid priority $\implies$ higher individual trading costs
- higher trading costs $\implies$ higher valuation required to trade profitably
- prices overshoot more, liquidity decreases

Literature

Blockchain & DEX microstructure

Equilibrium liquidity provision

Park (2023, *MS*) · Malinova–Park (2024) · Hasbrouck–Rivera–Saleh (2026, *MS*) · Lehar–Parlour (2025, *JF*)

LP losses & other markets

Milionis et al. (2022) · Hasbrouck–Rivera–Saleh (2023) · Cartea–Drissi–Monga (2023, 2024, *SIAM*)

Informed liquidity provision

Klein et al. (2023)

Blockchain infrastructure (gas fees)

Capponi–Jia–Yu (2024, *RFS*)

Others

Angeris–Evans–Chitra (2021) · Capponi–Iyengar–Sethuraman (2023, *F&T*)

Classical microstructure

Information monopolist — Kyle (1985, *Econometrica*)

Competition on information — Holden–Subrahmanyam (1992, *JF*), Foster–Viswanathan (1996, *JF*)

Multi-stage inventory models — Biais (1993, *JF*), De Futos–Manzano (2002, *JF*)

Compete for queue position (not speed) — O'Hara (1998, *book*)

Information and prices — Grossman–Stiglitz (1980, *AER*), Verrecchia (1982, *Econometrica*), Kyle (1989, *RES*), Baldauf–Mollner (2020, *JF*)

DEX $\approx$ parametrized price schedule in electronic markets — Glosten (1994, *JF*), Biais–Martimort–Rochet (2000, *Econometrica*)

Dealer markets / market making — Ho–Stoll (1983, *JF*), Glosten–Milgrom (1985, *JFE*)

Political contests (multi-prize)

Multi-prize contests — Lazear–Rosen (1981, *JPE*), Hillman–Riley (1989), Barut–Kovenock (1998), Klose–Kovenock (2015)

Multi-prize contests with a fixed prize vector — Moldovanu–Sela (2001, *AER*)

Market design - HFT

Periodic auctions — Madhavan (1992, *JF*), Budish–Cramton–Shim (2015, *QJE*)

Mollner & Baldauf (2020, *JF*), Menkveld & Zoican (2017, *RFS*)

Micro-burst fees on liquidity-taking orders — Brolley & Zoican (2023, *JFM*), Hosman et al. (2019)

Smart contract

Liquidity providers set liquidity L

- Mechanical price update rule: after a trade Q , the price updates by

$$P \longrightarrow P + \underbrace{2Q/L}_{\text{price impact}}$$

- Convex price schedule:

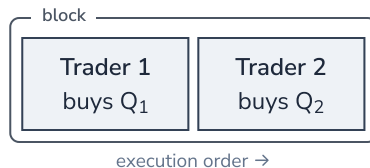
$$Q \times \left(P + \underbrace{Q/L}_{\text{execution cost per unit}} \right)$$

- Revenue to the liquidity provider:

$$\pi \times Q$$

Why compete?

- Assume $P = 0$
- Trader 1 buys Q_1 first, then trader 2 buys Q_2



- Trader 1 pays

$$Q_1 \times \left(\underbrace{Q_1/L + \pi}_{\text{exec. costs}} \right)$$

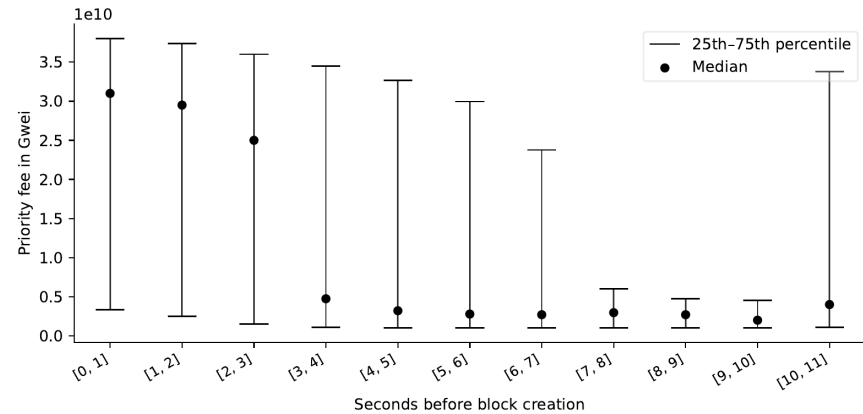
- Trader 2 pays

$$Q_2 \times \left(\underbrace{2Q_1/L}_{\text{impact from trader 1}} + \underbrace{Q_2/L + \pi}_{\text{exec. costs}} \right)$$

Competition structure

Informed flow arrives just before block creation

- Traders who submit high priority fees submit late
- Competition among informed traders is a blind single-round multi-prize auction



Priority fee distribution as a function of submission time over a blockchain slot

Data: Ethereum public mempool, ≈10 million transactions, 20–21 March 2024 (source: EthPandaOps)

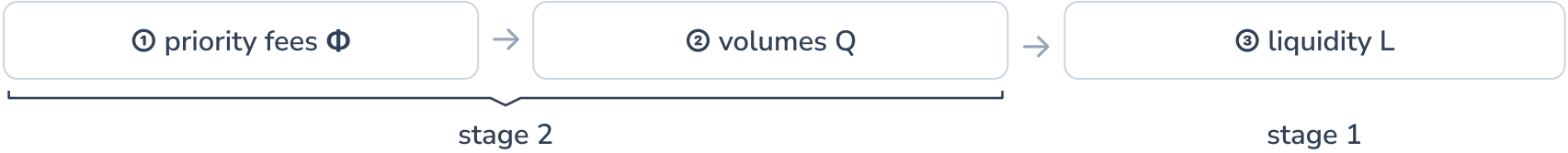
Model

A two-stage game

Given M informed traders



Solved backward



Stage 2: competition among informed traders

Setup

- Given M informed traders trading in the same direction (here, they buy)
- private valuations $v_i \in [0, \bar{v}] \sim F$; they set volumes Q_i and priority fees Φ_i

Expected wealth of trader i

$$\text{expected volume ahead of trader } i = \bar{Q}(\Phi_i) = \mathbb{E}_i \left[\underbrace{\sum_{j: \Phi_j > \Phi_i} Q_j}_{\text{volume ahead of } i} \right] \quad \text{expected price impact} = \Delta(\Phi_i) = \frac{2 \bar{Q}(\Phi_i)}{L}$$

$$\mathbb{E}_i[W_i] = \underbrace{Q_i v_i}_{\text{inventory}} - \underbrace{\Phi_i}_{\text{priority fee}} - Q_i \left(\underbrace{\frac{Q_i}{L} + \pi}_{\text{exec. costs}} + \underbrace{\Delta(\Phi_i)}_{\text{expected price impact}} \right)$$

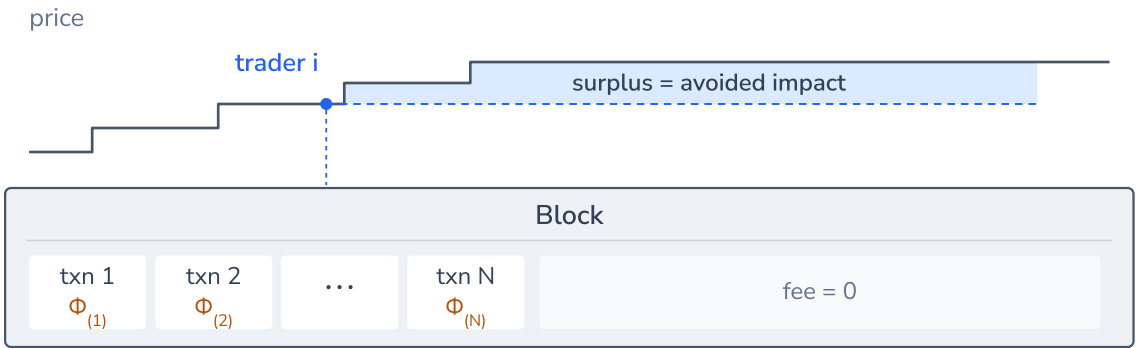
Objective

$$\sup_{\Phi_i, Q_i} \mathbb{E}_i[W_i] = \sup_{Q_i} \sup_{\Phi_i} \mathbb{E}_i[W_i]$$

① priority fees → ② volumes → ③ liquidity

- Fix a distribution of volumes $Q \in [0, \overline{Q}] \sim G$
- Trader i sets Φ_i by solving

$$\sup_{\Phi_i} \left\{ - \underbrace{\Phi_i}_{\text{priority fee}} + \underbrace{\frac{2Q_i}{L} \mathbb{E}_i \left[\sum_{j: \Phi_j < \Phi_i} Q_j \right]}_{\text{surplus = impact avoided}} \right\}$$

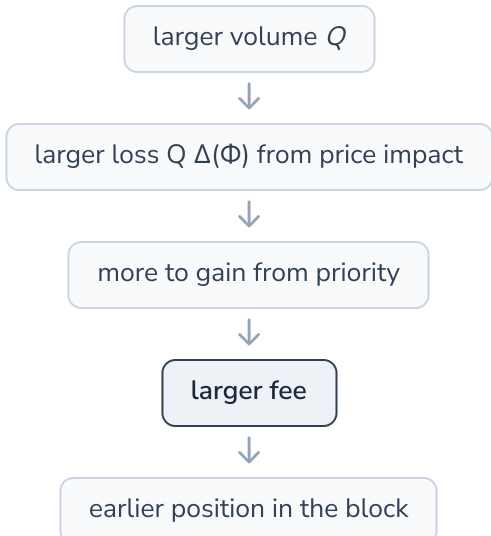


① priority fees → ② volumes → ③ liquidity

$$\Phi^*(Q_i) = \frac{2(M-1)}{L} \int_0^{Q_i} x^2 dG(x)$$

EQUILIBRIUM FEE

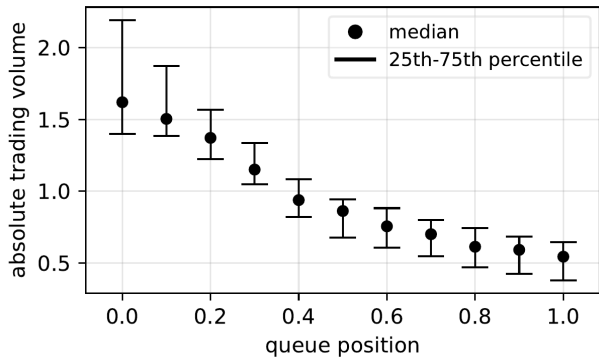
1. priority fees increase with volume Q_i



① priority fees → ② volumes → ③ liquidity

When price formation happens on-chain:

- absolute volume decreases with queue position



Distribution of normalized absolute volume as a function of queue position Data: 75 Uniswap v3 pools whose asset is *not* on Binance: 317,901 trades in 36,692 blocks, 2021–2026

① priority fees → ② volumes → ③ liquidity

$$\Phi^*(Q_i) = \frac{2(M-1)}{L} \int_0^{Q_i} x^2 dG(x)$$

EQUILIBRIUM FEE

2. priority fees increase with competition M

3. priority fees decrease with liquidity L



① priority fees → ② volumes → ③ liquidity

- Trader i anticipates the fee $\Phi^*(Q_i)$ and chooses Q_i :

$$\sup_{Q_i \geq 0} \left\{ \underbrace{Q_i v_i}_{\text{inventory}} - \underbrace{Q_i \left(\pi + \frac{Q_i}{L} \right)}_{\text{execution costs}} - \underbrace{\frac{2(M-1)}{L} \int_0^{Q_i} x^2 dG(x)}_{\text{priority fee}} - \underbrace{\frac{2Q_i(M-1)}{L} \int_{Q_i}^{\bar{Q}} x dG(x)}_{\text{expected price impact}} \right\}$$

- Equilibrium volume

$$Q^*(v_i) = \underbrace{L}_{\text{liquidity}} \times \frac{1}{2} \max \left\{ 0, \underbrace{(\bar{v} - \pi) e^{-(M-1)(1-F(v_i))}}_{\text{profit net of fees}} - \underbrace{\int_{v_i}^{\bar{v}} e^{-(M-1)(F(u)-F(v_i))} du}_{\text{impact of competing orders}} \right\}$$

① priority fees → ② volumes → ③ liquidity

EQUILIBRIUM VOLUME

- increasing in valuation v , liquidity L

ENDOGENOUS SELECTION

- only high-valuation traders trade: cutoff $\underline{v}(M)$
- cutoff rises with competition: $\lim_{M \rightarrow \infty} \underline{v}(M) = \bar{v}$

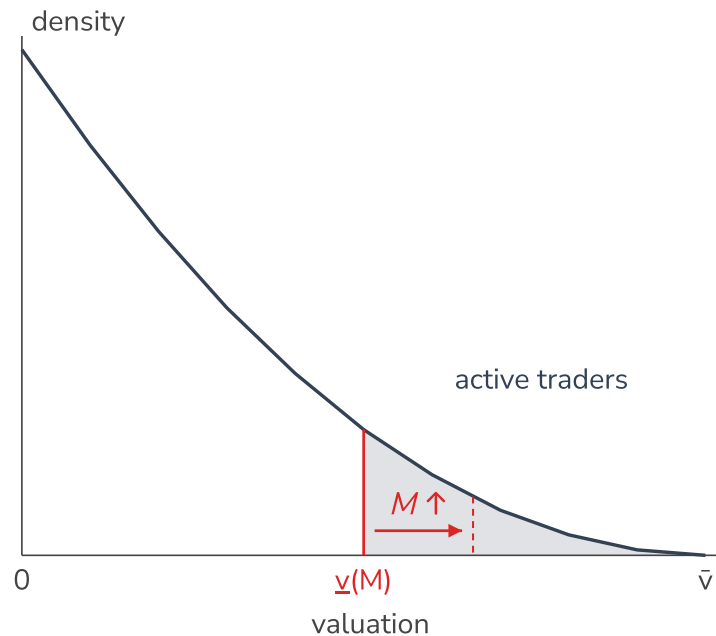


① priority fees → ② volumes → ③ liquidity

Consequences

- information is truncated
- price is biased

$$\lim_{M \rightarrow \infty} \underbrace{\frac{M(\underline{v}(M) - \pi)}{M - 1}}_{\text{end-of-block price}} = \bar{v} - \pi$$



① priority fees → ② volumes → ③ liquidity

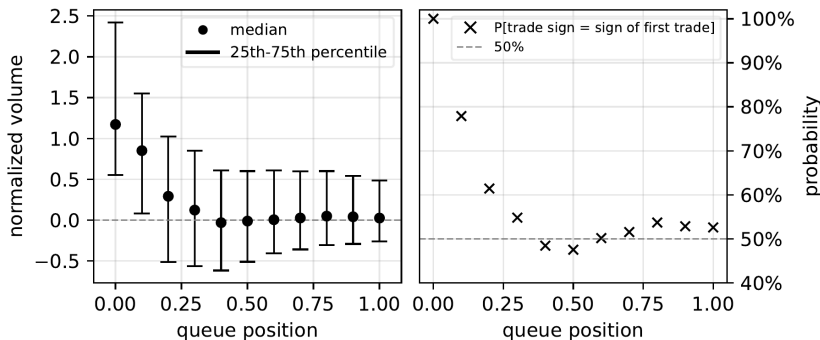
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Distribution of absolute one-block returns in bp (1 Jan 2021 – 31 Dec 2023)
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Equilibrium structure of a block predicted by the model

- two types of traders: **informed** (nonzero fees), **uninformed noise** (zero fees)
 - **front of block**: informed (same direction, decreasing volumes)
 - **back of block**: noise (two-sided, net volume ≈ 0)



Distribution of normalized volume as a function of queue position Data: 75 Uniswap v3 pools whose asset is *not* on Binance: 317,901 trades in 36,692 blocks, 2021–2026

Contrast with LOBs

One block interval spans **many trading rounds** in an electronic limit order book

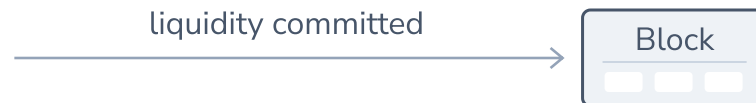
Electronic limit order book



trade → learn → revise quotes

- A lower-valuation trader can **post a limit sell order**, or sell at a bid considered high
- Orders contribute to **price discovery**
- Quotes and liquidity adjust as information arrives

Blockchain market: our model



information revealed at clearing

- Informed traders submit near **block creation**
- Trades execute sequentially against **passive liquidity**, with mechanical price updates
- Lower-valuation traders **optimally stay out**

Paid priority selects whose valuations enter the end-of-block price

① priority fees → ② volumes → ③ liquidity

liquidity provision is competitive

revenue from price-sensitive noise demand = adverse selection

EQUILIBRIUM LIQUIDITY

- liquidity decreases with the number of traders

M

(Holden & Subrahmanyam 1992)

- market viability condition

(Glosten & Milgrom 1985, Glosten 1989, Leach &

Madhavan 1989)

① priority fees → ② volumes → ③ liquidity → ④ entry M

We endogenize entry (M) .

- Before observing valuations, traders pay information cost C if expected ex-ante profit covers it

$$C \leq \underbrace{H(M)}_{\text{ex-ante profit per informed trader}}$$

- information paradox

$$\underbrace{C \downarrow}_{\text{cheaper info}} \Rightarrow \underbrace{M \uparrow}_{\text{more traders}} \Rightarrow \underbrace{v(M) \uparrow}_{\text{higher cutoff}} \Rightarrow \left\{ \begin{array}{l} \text{less information revealed} \\ \text{prices more biased} \end{array} \right.$$

- Blockchain markets can be improved (not in this paper)!
 - The length of a block can be tailored to improve market efficiency
 - Market design question with new types of agents (validators)

Block time: security / market efficiency

Pros

Block time: time to run **consensus**

- longer blocks → **security** and decentralisation

Cons

Block time: length of a trading round

- extreme valuations more likely
- noise trading decreases

A longer block amplifies every friction

- the participation cutoff increases, information is more truncated, prices are more extreme
- less liquidity, market shutdown more likely

Takeaway:

The features of decentralisation undermine blockchain markets

paid priority + ex-ante price update schedule + heterogeneous valuations = selection

Thank you!

Why rule out short selling?

Lower-valuation traders may want to sell after high-valuation buyers push prices up

Candidate equilibria:

1. Sellers pay priority fees:

- the seller with the lowest positive fee can deviate
- reduce priority fee, execute after buyers, and improve execution price

2. Groups with the same priority fee and nonzero expected net volume:

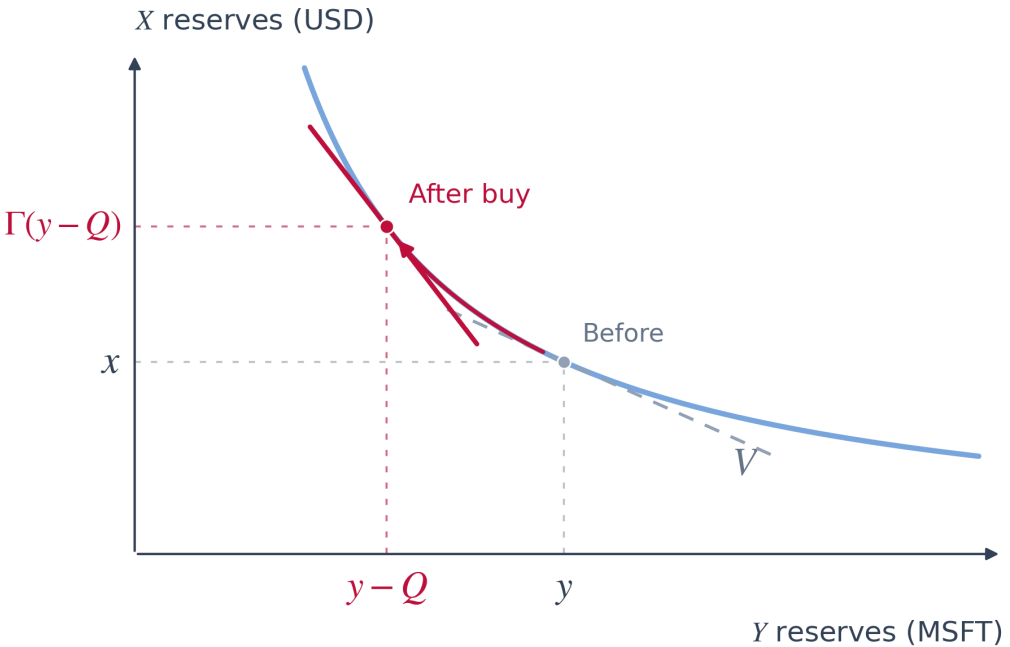
- net buying → a buyer can deviate
- net selling → a seller can deviate

3. Same fee, zero expected net volume:

- Possible equilibrium, but fragile: requires a group of buyers and sellers, with zero net volume, to coordinate on the same fee
- deviation with risk aversion: a tiny positive fee → same execution price but reduced uncertainty

AMMs

An automated market maker is a smart contract that prices trades from its reserves



After buying Q MSFT

$$V_{\text{after}} = -\Gamma'(y - Q)$$

$$V_{\text{after}} > V$$

MSFT reserves fall
The marginal price rises